

C01-AT3-Coding Challenge

1. Question



SIMATS Online Programming Lab

Java, Python, JavaScript



C01-AT3-Coding Challenge

View History

QUESTIONS

Q1

Smart University Payroll Management System (2 marks)

Q2

Smart University Payroll Management System (2 marks)

2/2 answered

Q1 Smart University Payroll Management System

25 marks

A university employs three categories of staff: Teaching faculty, Technical staff, and Visiting faculty. The payroll system should calculate the monthly salary based on the employee type while maintaining a common employee hierarchy. Requirements: Create a superclass Employee with the following private data members: employeeId, employeeName, department, basicSalary. Provide Default constructor, Parameterized constructor (better and better methods: displayEmployee(), calculateSalary()). Create the following subclasses: TeachingFaculty, TeachingInstructor, OverTimeInstructor, Salary, basicSalary, + (overTimeHours + (overTimeRate * overTimeHours)) calculateSalary(). Technical staff: OverTimeInstructor, OverTimeRate, Salary, basicSalary, + (overTimeHours + (overTimeRate * overTimeHours)) calculateSalary(). Visiting faculty: OverTimeInstructor, OverTimeRate, Salary, basicSalary, + (overTimeHours + (overTimeRate * overTimeHours)) calculateSalary(). In all subclasses, Create an array of Employee objects and demonstrate runtime polymorphism. Display Employee details. Monthly salary. Highest-paid employee. Average salary. Search an employee using the employee ID.

```

Test Case 1 (Partial Case)
Input
3
TeachingFaculty
101
Dr. Arun
0.18
30000
20
300
TechnicalStaff
102
Mr. Ravi
0.02
30000
15
300
VisitingFaculty
103
Dr. Kumar

```

1. Answer

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JAVA PROGRAMMING

Exit

Your Code

```

1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5         Scanner scanner = new Scanner(System.in);
6         int id=Integer.parseInt(sc.nextLine()).trim();
7         Employee[] emp=new Employee[5];
8         for(int i=0;i<emp.length;i++)
9             String type=emp.nextLine().trim();
10            while(type.isEmpty())
11                type=emp.nextLine().trim();
12            int id=Integer.parseInt(sc.nextLine()).trim();
13            String name=emp.nextLine().trim();
14            String department=emp.nextLine().trim();
15            double basicSalary=Double.parseDouble(sc.nextLine().trim());
16            int salary=Integer.parseInt(sc.nextLine().trim());
17            double value=Double.parseDouble(sc.nextLine().trim());
18            if(type.equalsIgnoreCase("TechnicalStaff"))
19                emp[i]=new TechnicalStaff(id,name,dept,basicSalary,value);
20            else if(type.equalsIgnoreCase("TechnicalStaff"))
21                emp[i]=new TechnicalStaff(id,name,dept,basicSalary,value);
22            else if(type.equalsIgnoreCase("WorkingFaculty"))
23                emp[i]=new WorkingFaculty(id,name,dept,basicSalary,value);
24            }
25        }
26        During scanner.nextLine().trim();
27        while(temp.isEmpty())
28            temp=emp.nextLine().trim();
29        emp.remove();
30        if(temp.equalsIgnoreCase("Search Employee ID"))
31            searchId=Integer.parseInt(sc.nextLine().trim());
32            searchId=Integer.parseInt(emp);
33        System.out.println("Employee Details");
34        System.out.println();
35        Employee highest=emp[0];
36        double total=0;
37        for(Employee emp1)
38            System.out.println("Employee ID : "+emp1.getId());
39            System.out.println("Name : "+emp1.getName());

```

Input

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99
100

```

Output

```

Employee Details
Employee ID : 123
Name : Dr. Ram
Department : CSE

```

2. Question



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JAVA PROGRAMMING

4. Display
10/03/2024

CO1-AT3-Coding Challenge
View Only

QUESTIONS

Q1
Single University Layout Management System
22 Marks

Q2
Smart Vehicle Insurance Claim System
22 Marks

2/2 answered

Smart Vehicle Insurance Claim System
22 Marks

Problem Statement An insurance company processes claims for three categories of vehicles: Car, Bike, and Truck. The claim amount depends on the vehicle type. Requirements: Superclass: Vehicle Private members: vehicleNumber (ownerName insuredAmount) Subclasses: Car, Bike, Truck Each subclass overrides calculateClaimAmount() using different formulas. The application should: Read details for N vehicles Store them using an array of superclass references Calculate claim amounts using runtime polymorphism Search by vehicle number Display the vehicle with the highest claim amount Question 3 (4x4=16) Online Shopping Order Processing System (30 Marks) **Problem Statement** An e-commerce company sells products in three categories: Electronics, Clothing, Grocery. The billing process varies for each category. Requirements: Superclass: Product Private members: productID, productName, price, quantity Subclasses: Electronics, Clothing, Grocery Each subclass overrides calculateBill(). The program should: Store products in an array of superclass references Calculate the final bill using runtime polymorphism Search products by ID Display the most expensive order Compute total revenue

Claim Formula (Recommended):

Car
Claim = insuredAmount * 0.80

Bike
Claim = insuredAmount * 0.60

Truck
Claim = insuredAmount * 0.90

Test Case 1

Input

3

Car
TH10AB1234
Ramesh
800000

Bike
TH00C15678
Suresh
100000

2. Answer



SIMATS Online Programming Lab
JAVA PROGRAMMING

4. Display
10/03/2024

Your Code

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc=new Scanner(System.in);
5         int n=Integer.parseInt(sc.nextLine().trim());
6         Vehicle[] v=new Vehicle[n];
7         for(int i=0;i<n;i++){
8             String temp=sc.nextLine().trim();
9             while(temp.isEmpty())
10                 temp=sc.nextLine().trim();
11             String number=sc.nextLine().trim();
12             String owner=sc.nextLine().trim();
13             double amount=Double.parseDouble(sc.nextLine().trim());
14             if (type.equalsIgnoreCase("Car"))
15                 v[i]=new Car(number,owner,amount);
16             else if (type.equalsIgnoreCase("Bike"))
17                 v[i]=new Bike(number,owner,amount);
18             else if (type.equalsIgnoreCase("Truck"))
19                 v[i]=new Truck(number,owner,amount);
20         }
21         String temp=sc.nextLine().trim();
22         while(temp.isEmpty())
23             temp=sc.nextLine().trim();
24         String search;
25         if (temp.equalsIgnoreCase("Search Vehicles"))
26             searched=sc.nextLine().trim();
27         else
28             searched=temp;
29         Vehicle highest=0;
30         for (Vehicle v1) {
31             System.out.println("Vehicle : "+v1.getVehicleNumber());
32             System.out.println("Claim Amount : "+v1.calculateClaimAmount());
33             System.out.println();
34             if (v1.calculateClaimAmount() > highest.calculateClaimAmount())
35                 highest=v1;
36         }
37         System.out.println("Highest Claim Vehicle");
38         System.out.println(highest.getVehicleNumber());
39         System.out.println(highest.calculateClaimAmount());
40         System.out.println();
41         boolean found=false;
42         for (Vehicle v1) {
43             if (v1.getVehicleNumber().equals(searched)) {
44                 System.out.println("Vehicle : "+v1.getVehicleNumber());
45                 System.out.println("Claim Amount : "+v1.calculateClaimAmount());
46                 System.out.println();
47                 found=true;
48             }
49         }
50         if (!found)
51             System.out.println("Vehicle not found");
52     }
53 }
```

Input

```
3
Car
TH10AB1234
Ramesh
800000
Bike
TH00C15678
Suresh
100000
```

Output

```
Vehicle : TH10AB1234
Claim Amount : 640000.0
Vehicle : TH00C15678
Claim Amount : 60000.0
```